KKT Conditions of the SOC-Based ISO Dispatch Problem and Their Equivalence with the Multi-Agent Complementarity Problem

Abstract

We derive in full the Karush–Kuhn–Tucker (KKT) system of the ISO clearing problem for state-of-charge (SOC) based storage participation, with the network, generation and demand constraints written out explicitly rather than collected into the compact operators Bx and $Cx \le d$; write the equilibrium problem of the market participants as a mixed complementarity problem (MCP); and prove that the two systems coincide. The ISO's dual variable on the SOC balance is shown to be an ISO-computed shadow price of stored energy obeying a backward Bellman recursion whose terminal condition is the end-of-horizon bid, and the nodal prices are shown to decompose into a reference price plus a shift-factor weighted sum of congestion prices. Consequently the ISO dispatch, together with the duals of the nodal balance, is a competitive equilibrium: no participant — including the storage owner, which submits no in-horizon energy offer — has a profitable unilateral deviation, and the settlement is revenue adequate with merchandising surplus exactly equal to congestion rent. Three conditions are necessary: concavity of the end-of-horizon bid, convexity of the feasible set, and the use of prices that are duals of the ISO problem rather than of a reduced clearing condition. We verify the whole system numerically on two cases: the single-node six-interval base case of the source paper, where the published price vector turns out not to be dual feasible, and a three-node, four-interval networked case with congestion, generator ramping and elastic demand.

1 Scope

References to the source paper [1] are written as "Eq. (4) of [1]"; all displayed equations of this note are numbered independently in the right margin. Problem (9) of [1] is stated compactly as

$$\min_{x,p,g,s} \Big\{ \Delta T c^\top x + \Delta T(\epsilon 1^\top p + \zeta 1^\top g) - b(s_T - \bar{s}_T) \;\Big|\; (1)\text{–}(7),\ p - g = Bx,\ Cx \le d \Big\}, \qquad (1)$$

in which the entire non-storage market — network, generators, loads — is compressed into the injection map B and the polyhedron $Cx \le d$. That compression is harmless for the existence argument but it hides exactly the objects a market model has to price: nodal balance, Kirchhoff's laws, thermal limits, offer blocks, ramp rates, and demand bids. Section 3 therefore expands (1) into its full disaggregated form, and everything that follows is derived from the expanded model. Section 3.1 records the mapping back to (1).

2 Notation

Sets and indices			
$t \in \mathcal{T} = \{1, \dots, T\}$	dispatch intervals, length ΔT (h)	$n \in \mathcal{N}$	buses (nodes)
$\ell \in \mathcal{L}$	transmission lines	$i \in \mathcal{G}$, $k \in \mathcal{K}_i$	generating units, offer blocks
$m \in \mathcal{D}$, $j \in \mathcal{J}_m$	loads, demand-bid blocks	$e \in \mathcal{E}$	storage units
$\mathcal{G}(n), \mathcal{D}(n), \mathcal{E}(n)$	units located at bus n	$n(i), n(m), n(e)$	bus of a unit
Primal variables			
$q_{i,k,t} \geq 0$	generation from block k (MW)	$d_{m,j,t} \geq 0$	served demand block (MW)
$f_{\ell,t}$	active power flow on line ℓ (MW)	$\theta_{n,t}$	voltage angle at bus n (rad)
$p_{e,t} \geq 0$	storage charge power (MW)	$g_{e,t} \geq 0$	storage discharge power (MW)
$s_{e,t}$	SOC at end of interval t (MWh)		
Data			
$c_{i,k}$	offer price of block k (\$/MWh)	$\bar{q}_{i,k}$	block size (MW)
$u_{m,j}$	demand bid price (\$/MWh)	$\bar{d}_{m,j}$	bid block size (MW)
$\bar{R}_i, \underline{R}_i$	up / down ramp limits (MW/interval)	y_ℓ	series susceptance (MW/rad)
$\bar{f}_\ell$	thermal limit (MW)	$A \in \mathbb{R}^{\lvert\mathcal{N}\rvert \times \lvert\mathcal{L}\rvert}$	bus–line incidence matrix
α_e, β_e	charge / inverse-discharge efficiency	γ_e	self-discharge retention factor
ϵ_e, ζ_e	charge / discharge degradation (\$/MWh)	$\bar{s}_e, \underline{s}_e$	SOC limits (MWh)
$\bar{p}_e, \bar{g}_e$	power limits (MW)	$b_e(\cdot), w_e(\cdot)$	end-of-horizon bid and its slope
$D_{n,t}$	fixed (price-inelastic) demand (MW)	$r \in \mathcal{N}$	reference bus

$A_{n\ell} = +1$ if n is the from-bus of ℓ, -1 if the to-bus, 0 otherwise; write $\mathrm{fr}(\ell)$ and $\mathrm{to}(\ell)$ for those buses. The storage recursion, Eq. (4) of [1], is

$$s_{e,t} = \gamma_e s_{e,t-1} + \Delta T\, \alpha_e p_{e,t} - \Delta T\, \beta_e g_{e,t}, \qquad e \in \mathcal{E},\ t \in \mathcal{T}, \tag{2}$$

with $s_{e,0}$ given, and the tightened SOC limits, Eqs. (6)–(7) of [1], are

$$\gamma_e s_{e,t-1} + \Delta T\, \alpha_e p_{e,t} \leq \bar{s}_e, \qquad \underline{s}_e - \gamma_e s_{e,t-1} + \Delta T\, \beta_e g_{e,t} \leq 0. \tag{3}$$

Substituting (2) into (3) gives the equivalent readable forms $s_{e,t} + \Delta T \beta_e g_{e,t} \leq \bar{s}_e$ and $s_{e,t} - \Delta T \alpha_e p_{e,t} \geq \underline{s}_e$, which is why (3) is tighter than bounding $s_{e,t}$ alone: it forbids trajectories that respect the limits at interval endpoints but violate them within the interval. The end-of-horizon bid, Eq. (8) of [1], is

$$b_e(\Delta) = \int_0^{\Delta} w_e(\sigma)\, d\sigma, \qquad \Delta := s_{e,T} - \bar{s}_{e,T}, \tag{4}$$

with $\bar{s}_{e,T}$ the reference terminal SOC.

Assumption 1 (Bid validity)**.** w_e is non-increasing for every $e \in \mathcal{E}$. Hence each b_e is concave and $-b_e$ convex.

Assumption 2 (Convex market)**.** The mutual-exclusivity condition, Eq. (3) of [1], is relaxed; no binary variables are present (Appendices A.2, A.3 of [1] excluded); losses are neglected or piecewise-linearised convexly; offer stacks are monotone, $c_{i,1} \leq c_{i,2} \leq \cdots$, and demand stacks are monotone non-increasing, $u_{m,1} \geq u_{m,2} \geq \cdots$.

Assumptions 1 and 2 are logically independent: the first concerns the *curvature of a function*, the second the *shape of a set*. Both are needed, and they fail for different reasons (Section 9).

3 The ISO problem in disaggregated form

$$\begin{aligned} \min_{q,d,f,\theta,p,g,s} \quad & \Delta T \sum_{t \in \mathcal{T}} \left[\sum_{i \in \mathcal{G}} \sum_{k \in \mathcal{K}_i} c_{i,k}\, q_{i,k,t} - \sum_{m \in \mathcal{D}} \sum_{j \in \mathcal{J}_m} u_{m,j}\, d_{m,j,t} + \sum_{e \in \mathcal{E}} \big(\epsilon_e p_{e,t} + \zeta_e g_{e,t} \big) \right] \\ & - \sum_{e \in \mathcal{E}} b_e\big(s_{e,T} - \bar{s}_{e,T} \big) \end{aligned} \tag{5}$$

subject to, for all $t \in \mathcal{T}$, with multipliers on the right:

$$\sum_{m\in\mathcal{D}(n)}\sum_{j} d_{m,j,t} + D_{n,t} + \sum_{\ell\in\mathcal{L}} A_{n\ell} f_{\ell,t} - \sum_{i\in\mathcal{G}(n)}\sum_{k} q_{i,k,t} - \sum_{e\in\mathcal{E}(n)} \big(g_{e,t} - p_{e,t}\big) = 0, \qquad \Delta T\lambda_{n,t} \text{ free} \tag{D1}$$

$$y_\ell\big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t}\big) - f_{\ell,t} = 0, \qquad \Delta T\kappa_{\ell,t} \text{ free} \tag{D2}$$

$$f_{\ell,t} - \bar{f}_\ell \le 0, \qquad -\bar{f}_\ell - f_{\ell,t} \le 0, \qquad \Delta T\bar{\mu}_{\ell,t},\ \Delta T\underline{\mu}_{\ell,t} \ge 0 \tag{D3}$$

$$\theta_{r,t} = 0, \qquad \Delta T\xi_t \text{ free} \tag{D4}$$

$$-q_{i,k,t} \le 0, \qquad q_{i,k,t} - \bar{q}_{i,k} \le 0, \qquad \Delta T\underline{\delta}_{i,k,t},\ \Delta T\bar{\delta}_{i,k,t} \ge 0 \tag{D5}$$

$$\sum_k q_{i,k,t} - \sum_k q_{i,k,t-1} - \bar{R}_i \le 0, \qquad \Delta T\bar{\tau}_{i,t} \ge 0 \tag{D6}$$

$$\sum_k q_{i,k,t-1} - \sum_k q_{i,k,t} - \underline{R}_i \le 0, \qquad \Delta T\underline{\tau}_{i,t} \ge 0 \tag{D7}$$

$$-d_{m,j,t} \le 0, \qquad d_{m,j,t} - \bar{d}_{m,j} \le 0, \qquad \Delta T\underline{\psi}_{m,j,t},\ \Delta T\bar{\psi}_{m,j,t} \ge 0 \tag{D8}$$

$$s_{e,t} - \gamma_e s_{e,t-1} - \Delta T\alpha_e p_{e,t} + \Delta T\beta_e g_{e,t} = 0, \qquad \pi_{e,t} \text{ free} \tag{D9}$$

$$\gamma_e s_{e,t-1} + \Delta T\alpha_e p_{e,t} - \bar{s}_e \le 0, \qquad \Delta T\bar{\nu}_{e,t} \ge 0 \tag{D10}$$

$$\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T\beta_e g_{e,t} \le 0, \qquad \Delta T\underline{\nu}_{e,t} \ge 0 \tag{D11}$$

$$-p_{e,t} \le 0, \quad p_{e,t} - \bar{p}_e \le 0, \qquad \Delta T\underline{\rho}_{e,t},\ \Delta T\bar{\rho}_{e,t} \ge 0 \tag{D12}$$

$$-g_{e,t} \le 0, \quad g_{e,t} - \bar{g}_e \le 0, \qquad \Delta T\underline{\sigma}_{e,t},\ \Delta T\bar{\sigma}_{e,t} \ge 0 \tag{D13}$$

for all $n \in \mathcal{N}$, $\ell \in \mathcal{L}$, $i \in \mathcal{G}$, $k \in \mathcal{K}_i$, $m \in \mathcal{D}$, $j \in \mathcal{J}_m$, $e \in \mathcal{E}$; (D6)–(D7) for $t \ge 2$.

Three modelling points. First, (D1) is written as *net withdrawal* $= 0$ so that $\lambda_{n,t}$ comes out as a price rather than its negative; the factors ΔT are inserted so that every multiplier carries units of \$/MWh. Second, (D2) is the DC linearisation of Kirchhoff's voltage law, and it is what makes the network more than a transportation problem: without it, flows could be routed freely and the price decomposition of Proposition 3 would collapse. Third, (D6)–(D7) give thermal units an *intertemporal* constraint of their own, so the comparison between the storage costate and the ramp wedge in Section 5 is available.

3.1 Mapping back to the compact form

Setting $x := \big(q, d, f, \theta\big)$, the compact objects of (1) are

$$c^\top x := \sum_{i,k,t} c_{i,k} q_{i,k,t} - \sum_{m,j,t} u_{m,j} d_{m,j,t}, \qquad (Bx)_{n,t} := \sum_{i\in\mathcal{G}(n),k} q_{i,k,t} - \sum_{m\in\mathcal{D}(n),j} d_{m,j,t} - D_{n,t} - \sum_{\ell} A_{n\ell} f_{\ell,t}, \tag{6}$$

so that (D1) is $p_{n,t} - g_{n,t} = (Bx)_{n,t}$ aggregated over $\mathcal{E}(n)$, and $Cx \le d$ collects (D3) and (D5)–(D8), with (D2) and (D4) as the equality rows of the same block. In the notation of [1] the multiplier μ on $Cx \le d$ is $\mu = \big(\bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}\big)$, of which $\mu_s := (\bar{\mu}, \underline{\mu})$ are the system-wide (transmission) components and the remainder are firm-private.

4 KKT system of the ISO problem

The Lagrangian is

$$\begin{aligned}\mathcal{L} = (5) &+ \Delta T \sum_{n,t} \lambda_{n,t}\Big[\textstyle\sum_{m,j} d_{m,j,t} + D_{n,t} + \sum_{\ell} A_{n\ell} f_{\ell,t} - \sum_{i,k} q_{i,k,t} - \sum_{e}(g_{e,t} - p_{e,t})\Big] \\ &+ \Delta T \sum_{\ell,t} \kappa_{\ell,t}\Big[y_\ell\big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t}\big) - f_{\ell,t}\Big] + \Delta T \sum_{\ell,t}\Big[\bar{\mu}_{\ell,t}(f_{\ell,t} - \bar{f}_\ell) + \underline{\mu}_{\ell,t}(-\bar{f}_\ell - f_{\ell,t})\Big] \\ &+ \Delta T \sum_{t} \xi_t \theta_{r,t} + \Delta T \sum_{i,k,t}\big[\bar{\delta}_{i,k,t}(q_{i,k,t} - \bar{q}_{i,k}) - \underline{\delta}_{i,k,t} q_{i,k,t}\big] \\ &+ \Delta T \sum_{i,t\geq 2}\Big[\bar{\tau}_{i,t}\big(Q_{i,t} - Q_{i,t-1} - \bar{R}_i\big) + \underline{\tau}_{i,t}\big(Q_{i,t-1} - Q_{i,t} - \underline{R}_i\big)\Big] \\ &+ \Delta T \sum_{m,j,t}\big[\bar{\psi}_{m,j,t}(d_{m,j,t} - \bar{d}_{m,j}) - \underline{\psi}_{m,j,t} d_{m,j,t}\big] \;+\; \mathcal{L}^{\mathrm{stor}},\end{aligned} \tag{7}$$

where $Q_{i,t} := \sum_k q_{i,k,t}$ and

$$\begin{aligned}\mathcal{L}^{\mathrm{stor}} = \sum_{e,t}\Big\{&\pi_{e,t}\big[s_{e,t} - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} + \Delta T \beta_e g_{e,t}\big] + \Delta T \bar{\nu}_{e,t}\big[\gamma_e s_{e,t-1} + \Delta T \alpha_e p_{e,t} - \bar{s}_e\big] \\ &+ \Delta T \underline{\nu}_{e,t}\big[\underline{s}_e - \gamma_e s_{e,t-1} + \Delta T \beta_e g_{e,t}\big] + \Delta T\big[\bar{\rho}_{e,t}(p_{e,t} - \bar{p}_e) - \underline{\rho}_{e,t} p_{e,t} \\ &+ \bar{\sigma}_{e,t}(g_{e,t} - \bar{g}_e) - \underline{\sigma}_{e,t} g_{e,t}\big]\Big\}.\end{aligned} \tag{8}$$

The multiplier $\pi_{e,t}$ on (D9) is the *marginal value of stored energy* at the end of interval t (\$/MWh of SOC); it does all the economic work below.

4.1 Stationarity

Generation $q_{i,k,t}$. $\partial\mathcal{L}/\partial q_{i,k,t} = \Delta T\big[c_{i,k} - \lambda_{n(i),t} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + (\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})\big] = 0$, i.e.

$$\lambda_{n(i),t} = c_{i,k} + \bar{\delta}_{i,k,t} - \underline{\delta}_{i,k,t} + \underbrace{(\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})}_{=:\,\varrho_{i,t}\ \text{(ramp wedge)}}, \tag{K1a}$$

with the convention $\bar{\tau}_{i,1} = \underline{\tau}_{i,1} = \bar{\tau}_{i,T+1} = \underline{\tau}_{i,T+1} = 0$.

Demand $d_{m,j,t}$. $\partial\mathcal{L}/\partial d_{m,j,t} = \Delta T\big[-u_{m,j} + \lambda_{n(m),t} + \bar{\psi}_{m,j,t} - \underline{\psi}_{m,j,t}\big] = 0$, i.e.

$$\lambda_{n(m),t} = u_{m,j} - \bar{\psi}_{m,j,t} + \underline{\psi}_{m,j,t}. \tag{K1b}$$

Flows $f_{\ell,t}$. The term $A^\top\lambda$ enters from (D1):

$$\kappa_{\ell,t} = \big(\lambda_{\mathrm{fr}(\ell),t} - \lambda_{\mathrm{to}(\ell),t}\big) + \big(\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}\big) = \big(A^\top\lambda_t\big)_\ell + \nu_{\ell,t}, \qquad \nu_{\ell,t} := \bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}. \tag{K1c}$$

Angles $\theta_{n,t}$. Only (D2) and (D4) contain θ:

$$\big(A\,\mathrm{diag}(y)\,\kappa_t\big)_n + \xi_t\,\mathbf{1}\{n = r\} = 0 \qquad\Longleftrightarrow\qquad A\,\mathrm{diag}(y)\,\kappa_t = -\xi_t e_r. \tag{K1d}$$

Summing (K1d) over n and using $\mathbf{1}^\top A = 0$ gives $\xi_t = 0$: the reference angle is a normalisation and carries no economic content. Hence $A\,\mathrm{diag}(y)\,\kappa_t = 0$, which is the dual form of Kirchhoff's voltage law.

Charging $p_{e,t}$.

$$\lambda_{n(e),t} + \epsilon_e = \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big) + \underline{\rho}_{e,t} - \bar{\rho}_{e,t}. \tag{K2}$$

Discharging $g_{e,t}$.

$$\lambda_{n(e),t} = \zeta_e + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) - \underline{\sigma}_{e,t} + \bar{\sigma}_{e,t}. \tag{K3}$$

SOC $s_{e,t}$, $t = 1, \dots, T-1$.

$$\pi_{e,t} = \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big). \tag{K4}$$

Terminal SOC $s_{e,T}$.

$$\pi_{e,T} = w_e\big(s_{e,T} - \bar{s}_{e,T}\big), \qquad \text{or} \;\; \pi_{e,T} \in \partial b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \text{ for piecewise-linear } b_e. \tag{K5}$$

4.2 Dual feasibility and complementary slackness

All of $\bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}, \bar{\nu}, \underline{\nu}, \bar{\rho}, \underline{\rho}, \bar{\sigma}, \underline{\sigma} \geq 0$ (K6), and

$$0 \leq \bar{\mu}_{\ell,t} \;\perp\; \bar{f}_\ell - f_{\ell,t} \geq 0, \qquad 0 \leq \underline{\mu}_{\ell,t} \;\perp\; f_{\ell,t} + \bar{f}_\ell \geq 0, \tag{K7}$$

$$0 \leq \underline{\delta}_{i,k,t} \;\perp\; q_{i,k,t} \geq 0, \qquad 0 \leq \bar{\delta}_{i,k,t} \;\perp\; \bar{q}_{i,k} - q_{i,k,t} \geq 0, \tag{K8}$$

$$0 \leq \bar{\tau}_{i,t} \;\perp\; \bar{R}_i - Q_{i,t} + Q_{i,t-1} \geq 0, \qquad 0 \leq \underline{\tau}_{i,t} \;\perp\; \underline{R}_i - Q_{i,t-1} + Q_{i,t} \geq 0, \tag{K9}$$

$$0 \leq \underline{\psi}_{m,j,t} \;\perp\; d_{m,j,t} \geq 0, \qquad 0 \leq \bar{\psi}_{m,j,t} \;\perp\; \bar{d}_{m,j} - d_{m,j,t} \geq 0, \tag{K10}$$

$$0 \leq \bar{\nu}_{e,t} \;\perp\; \bar{s}_e - \gamma_e s_{e,t-1} - \Delta T \alpha_e p_{e,t} \geq 0, \qquad 0 \leq \underline{\nu}_{e,t} \;\perp\; \gamma_e s_{e,t-1} - \Delta T \beta_e g_{e,t} - \underline{s}_e \geq 0, \tag{K11}$$

$$0 \leq \underline{\rho}_{e,t} \;\perp\; p_{e,t} \geq 0, \qquad 0 \leq \bar{\rho}_{e,t} \;\perp\; \bar{p}_e - p_{e,t} \geq 0,$$

$$0 \leq \underline{\sigma}_{e,t} \;\perp\; g_{e,t} \geq 0, \qquad 0 \leq \bar{\sigma}_{e,t} \;\perp\; \bar{g}_e - g_{e,t} \geq 0, \tag{K12}$$

together with primal feasibility (D1)–(D13). Here $a \perp b$ means $a^\top b = 0$.

4.3 Reduced form: eliminating the bound multipliers

Since $\underline{\rho}_{e,t}$ and $\bar{\rho}_{e,t}$ cannot both be positive, (K2) with (K12) is equivalent to the *box complementarity* statement

$$0 \leq p_{e,t} \leq \bar{p}_e \quad \perp \quad F^p_{e,t} := \lambda_{n(e),t} + \epsilon_e - \alpha_e\big(\pi_{e,t} - \bar{\nu}_{e,t}\big), \tag{K2$'$}$$

read as

$$p_{e,t} = 0 \Rightarrow F^p_{e,t} \geq 0, \qquad 0 < p_{e,t} < \bar{p}_e \Rightarrow F^p_{e,t} = 0, \qquad p_{e,t} = \bar{p}_e \Rightarrow F^p_{e,t} \leq 0, \tag{9}$$

and likewise

$$0 \leq g_{e,t} \leq \bar{g}_e \;\perp\; F^g_{e,t} := \zeta_e + \beta_e\big(\pi_{e,t} + \underline{\nu}_{e,t}\big) - \lambda_{n(e),t}, \qquad 0 \leq q_{i,k,t} \leq \bar{q}_{i,k} \;\perp\; c_{i,k} + \varrho_{i,t} - \lambda_{n(i),t}, \tag{K3$'$}$$

$$0 \leq d_{m,j,t} \leq \bar{d}_{m,j} \;\perp\; \lambda_{n(m),t} - u_{m,j}, \qquad -\bar{f}_\ell \leq f_{\ell,t} \leq \bar{f}_\ell \;\perp\; \kappa_{\ell,t} - \big(A^\top \lambda_t\big)_\ell. \tag{K1$'$}$$

The middle case of (9) is the operative one for pricing: *whenever a resource operates strictly interior, its own stationarity condition pins λ at its bus exactly*, leaving no clearing range. Section 8 exploits this.

4.4 Nodal price decomposition

Proposition 3 (LMP decomposition). *Let* $B_{\text{bus}} := A\,\mathrm{diag}(y)A^\top$*, let* $\mathcal{N}_{-r} := \mathcal{N} \setminus \{r\}$*, and let the shift-factor (PTDF) matrix relative to reference bus* r *be* $\Psi := \mathrm{diag}(y)\,A_{-r}^\top B_{-r}^{-1}$*, where subscript* $-r$ *deletes the row and column of* r*. Then* (K1c)–(K1d) *imply*

$$\lambda_{n,t} = \lambda_{r,t} - \sum_{\ell \in \mathcal{L}} \Psi_{\ell n}\big(\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t}\big), \qquad n \in \mathcal{N}. \tag{10}$$

Proof. Substituting (K1c) into (K1d) with $\xi_t = 0$ gives $A\,\mathrm{diag}(y)\big(A^\top\lambda_t + \nu_t\big) = 0$, i.e. $B_{\text{bus}}\lambda_t = -A\,\mathrm{diag}(y)\,\nu_t$. Since $B_{\text{bus}}\mathbf{1} = 0$, write $\lambda_t = \lambda_{r,t}\mathbf{1} + \tilde{\lambda}_t$ with $\tilde{\lambda}_{r,t} = 0$; deleting row r leaves $B_{-r}\tilde{\lambda}_{-r,t} = -A_{-r}\mathrm{diag}(y)\nu_t$, hence $\tilde{\lambda}_{-r,t} = -\big(\mathrm{diag}(y)A_{-r}^\top B_{-r}^{-1}\big)^\top \nu_t = -\Psi^\top\nu_t$, which is (10). □

So the energy component of every LMP is the reference-bus price, common to all buses and to all storage units, while the congestion component is the only source of locational difference. Storage at a congested bus therefore faces a price that already contains the network's scarcity, and (K2)–(K5) are stated at that price; no separate congestion term appears in the storage conditions.

5 Economic content

5.1 The SOC price is a Bellman costate

Conditions (K4)–(K5) form a backward recursion whose terminal condition is supplied by the participant's bid and which the ISO propagates backwards:

$$\pi_{e,T} = w_e(s_{e,T} - \bar{s}_{e,T}), \qquad \pi_{e,t} = \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big), \quad t = T-1,\dots,1. \tag{11}$$

This is the discrete costate (adjoint) equation of the storage dynamics, the exact analogue of a Bellman recursion for a value-to-go. Each term reads: γ_e discounts stored value for self-discharge; $\bar{\nu}_{e,t+1} > 0$ (tank full) *reduces* $\pi_{e,t}$, since energy carried forward cannot be stored; $\underline{\nu}_{e,t+1} > 0$ (tank empty) *raises* it, since energy carried forward relieves a binding floor. Solving forward from T,

$$\pi_{e,t} = \gamma_e^{T-t}\,w_e(s_{e,T} - \bar{s}_{e,T}) + \sum_{k=t+1}^{T} \gamma_e^{k-t}\big(\underline{\nu}_{e,k} - \bar{\nu}_{e,k}\big). \tag{12}$$

No forecast of λ by the storage owner appears anywhere: the entire intertemporal opportunity cost is reconstructed by the ISO from the single scalar function w_e.

5.2 The ramp wedge is the thermal analogue — and it is weaker

Condition (K1a) contains the corresponding object for a thermal unit, $\varrho_{i,t} = (\bar{\tau}_{i,t} - \underline{\tau}_{i,t}) - (\bar{\tau}_{i,t+1} - \underline{\tau}_{i,t+1})$. It is a *first difference* of ramp multipliers: it couples t only to $t \pm 1$, and it decays to zero the moment no ramp constraint binds. The storage costate (12) is a γ_e-discounted *sum* over the whole remaining horizon and is generically non-zero even when no SOC bound binds, because it inherits w_e at T. That is the formal sense in which energy-limited resources are intertemporal in a way ramp-limited ones are not, and it is why storage needs a terminal bid whereas a thermal unit does not.

5.3 Arbitrage conditions

If storage e charges strictly interior in t and discharges strictly interior in $t' > t$, then (K2′)–(K3′) give $\lambda_{n(e),t} + \epsilon_e = \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t})$ and $\lambda_{n(e),t'} = \zeta_e + \beta_e(\pi_{e,t'} + \underline{\nu}_{e,t'})$. Eliminating π with (11),

$$\boxed{\frac{\lambda_{n(e),t} + \epsilon_e}{\alpha_e} + \bar{\nu}_{e,t} = \gamma_e^{t'-t}\,\frac{\lambda_{n(e),t'} - \zeta_e}{\beta_e} + \sum_{k=t+1}^{t'-1} \gamma_e^{k-t}\big(\underline{\nu}_{e,k} - \bar{\nu}_{e,k}\big) - \gamma_e^{t'-t}\,\bar{\nu}_{e,t'}} \tag{13}$$

If no SOC bound binds on $(t, t']$ and $\gamma_e = 1$, (13) collapses to

$$\lambda_{n(e),t'} = \frac{\beta_e}{\alpha_e}\big(\lambda_{n(e),t} + \epsilon_e\big) + \zeta_e = \frac{1}{\eta^e_{\mathrm{RT}}}\big(\lambda_{n(e),t} + \epsilon_e\big) + \zeta_e : \tag{14}$$

the discharge price must exceed the charge price by the round-trip efficiency factor plus both degradation terms. Since (14) with $t = t'$ is infeasible for $\epsilon_e, \zeta_e \geq 0$ and $\beta_e \geq \alpha_e$, simultaneous charge and discharge never occurs at equilibrium prices, which is why Eq. (3) of [1] can be relaxed.

6 The participants' problems as a complementarity problem

6.1 Lagrangian decomposition

Dualise only the nodal balance (D1) in (7). The remainder separates additively across agents:

$$\mathcal{L} = \sum_{i\in\mathcal{G}} \mathcal{L}_i + \sum_{m\in\mathcal{D}} \mathcal{L}_m + \sum_{e\in\mathcal{E}} \mathcal{L}_e + \mathcal{L}_{\mathrm{net}} + \Delta T \sum_{n,t} \lambda_{n,t} D_{n,t}, \tag{15}$$

because no constraint other than (D1) couples variables belonging to different agents: (D5)–(D7) involve only unit i, (D8) only load m, (D9)–(D13) only storage e, and (D2)–(D4) only the network variables (f, θ). Each block is, up to sign, the objective of one agent's own profit maximisation at price λ. This is the whole of the equivalence result; the rest is bookkeeping.

6.2 Generator i

$$\max_{q_{i,\cdot,\cdot}\geq 0} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(i),t} Q_{i,t} - \sum_k c_{i,k} q_{i,k,t}\Big] \quad \text{s.t. } q_{i,k,t} \leq \bar{q}_{i,k}, \quad \big|Q_{i,t} - Q_{i,t-1}\big| \leq \{\bar{R}_i, \underline{R}_i\}, \tag{16}$$

whose KKT system is (K1a) with (K8)–(K9). Note that ramping makes this a genuinely multi-interval problem, so the generator is a price taker *over the whole horizon*, exactly as the storage is.

6.3 Load m

$$\max_{0\leq d_{m,j,t}\leq \bar{d}_{m,j}} \ \Delta T \sum_{t\in\mathcal{T}} \sum_j \big(u_{m,j} - \lambda_{n(m),t}\big) d_{m,j,t}, \tag{17}$$

with KKT system (K1b) and (K10).

6.4 Storage owner e (price taker)

$$\max_{p_e, g_e, s_e} \ \Delta T \sum_{t\in\mathcal{T}} \Big[\lambda_{n(e),t}\big(g_{e,t} - p_{e,t}\big) - \epsilon_e p_{e,t} - \zeta_e g_{e,t}\Big] + b_e\big(s_{e,T} - \bar{s}_{e,T}\big) \tag{18}$$

subject to the SOC dynamics (2), the SOC limits (3), and $0 \leq p_{e,t} \leq \bar{p}_e$, $0 \leq g_{e,t} \leq \bar{g}_e$. The objective is discharge revenue, less charging payment, less degradation, plus the declared value of the terminal SOC position. Attaching the same multiplier names to the same constraints, its KKT system is (K2), (K3), (K4), (K5) with (K11)–(K12), *verbatim*.

6.5 The network

The variables (f, θ) belong to no commercial participant; in an ISO market the system operator holds them. They can nevertheless be assigned to a fictitious price-taking *merchant transmission operator* that buys at each from-bus and sells at each to-bus,

$$\max_{f,\theta} \ -\Delta T \sum_{t\in\mathcal{T}} f_t^\top A^\top \lambda_t \quad \text{s.t.} \quad f_{\ell,t} = y_\ell\big(\theta_{\mathrm{fr}(\ell),t} - \theta_{\mathrm{to}(\ell),t}\big), \quad |f_{\ell,t}| \leq \bar{f}_\ell, \quad \theta_{r,t} = 0, \tag{19}$$

whose stationarity conditions in f and θ are exactly (K1c) and (K1d). This is a modelling device, not an institutional claim: the point is that the network block *does* decentralise, but only for an agent constrained to physically consistent flows. An agent that could route power freely subject to thermal limits alone would satisfy (K1c) but not (K1d), and the equivalence would fail — which is the formal statement of why loop flow cannot be traded bilaterally.

6.6 The multi-agent complementarity problem

Definition 4 (MCP)**.** Find $z = (q, d, f, \theta, p, g, s; \lambda, \kappa, \pi, \bar{\mu}, \underline{\mu}, \bar{\delta}, \underline{\delta}, \bar{\tau}, \underline{\tau}, \bar{\psi}, \underline{\psi}, \bar{\nu}, \underline{\nu})$ such that, for all $t \in \mathcal{T}, n \in \mathcal{N}, \ell \in \mathcal{L}, i \in \mathcal{G}, k, m \in \mathcal{D}, j, e \in \mathcal{E}$:

(a) Generators:	$0 \le q_{i,k,t} \le \bar{q}_{i,k} \;\perp\; c_{i,k} + \varrho_{i,t} - \lambda_{n(i),t}$, with (K9)	(20)
(b) Loads:	$0 \le d_{m,j,t} \le \bar{d}_{m,j} \;\perp\; \lambda_{n(m),t} - u_{m,j}$	(21)
(c) Storage, charge:	$0 \le p_{e,t} \le \bar{p}_e \;\perp\; \lambda_{n(e),t} + \epsilon_e - \alpha_e(\pi_{e,t} - \bar{\nu}_{e,t})$	(22)
(d) Storage, discharge:	$0 \le g_{e,t} \le \bar{g}_e \;\perp\; \zeta_e + \beta_e(\pi_{e,t} + \underline{\nu}_{e,t}) - \lambda_{n(e),t}$	(23)
(e) Storage, SOC:	$\pi_{e,t} - \gamma_e\big(\pi_{e,t+1} - \bar{\nu}_{e,t+1} + \underline{\nu}_{e,t+1}\big) = 0 \;(t < T), \quad \pi_{e,T} \in \partial b_e$	(24)
(f) SOC limits, dynamics:	(K11) and (2)	(25)
(g) Network, flows:	$-\bar{f}_\ell \le f_{\ell,t} \le \bar{f}_\ell \;\perp\; \kappa_{\ell,t} - \big(A^\top \lambda_t\big)_\ell$	(26)
(h) Network, angles:	$A\,\mathrm{diag}(y)\,\kappa_t = 0, \qquad \theta_{r,t} = 0, \qquad f_{\ell,t} = y_\ell\big(A^\top \theta_t\big)_\ell$	(27)
(i) Market clearing:	(D1) $\quad\perp\quad$ $\lambda_{n,t}$ free	(28)

Definition 5 (Competitive equilibrium)**.** $(q^*, d^*, f^*, \theta^*, p^*, g^*, s^*, \lambda^*)$ is a competitive equilibrium if q_i^* solves (16) for every i, d_m^* solves (17) for every m, (p_e^*, g_e^*, s_e^*) solves (18) for every e, (f^*, θ^*) solves (19), all at prices λ^*, and the nodal balance (D1) holds.

7 Equivalence and the equilibrium theorem

Theorem 6 (Equivalence of the KKT system and the MCP)**.** *Let Assumption 2 hold. Then z satisfies the ISO KKT system* (K1a)–(K12) *together with primal feasibility* (D1)–(D13) *if and only if z solves the MCP of Definition 4.*

Proof. ($\Rightarrow$) Partition the KKT conditions by the variable each is stationary in. (K1a) with (K8)–(K9) is (20), by the box-complementarity equivalence (K2′)–(9) applied to q; (K1b) with (K10) is (21); (K2) with (K12) is (22); (K3) with (K12) is (23); (K4)–(K5) is (24); (K11) and (D9) give (25); (K1c) with (K7) is (26); (K1d) with $\xi_t = 0$, together with primal feasibility of (D2) and (D4), is (27); and (D1) is (28). Every KKT condition is used exactly once and none is left over, because the Lagrangian separates as in (15).

($\Leftarrow$) Each step above is an identity between the same expressions, so the argument reverses term by term: the union of the agents' first-order conditions and market clearing reconstructs (K1a)–(K12) with the same multipliers. □

The content of Theorem 6 is that the ISO's KKT system is not merely *consistent* with the participants' conditions — it is *partitioned* by them. The blocks are disjoint and exhaustive, and the only condition belonging to no participant is market clearing, which is the ISO's own.

Theorem 7 (Equilibrium)**.** *Let Assumptions 1 and 2 hold and let $(q^*, d^*, f^*, \theta^*, p^*, g^*, s^*)$ solve* (5) *with multipliers $\lambda^*, \ldots$ Then it is a competitive equilibrium in the sense of Definition 5. Conversely, every competitive equilibrium yields an optimal solution of* (5)*.*

Proof. Under Assumption 1 each $-b_e$ is convex, so (5) minimises a convex objective; under Assumption 2 all constraints (D1)–(D13) are affine, so Abadie's constraint qualification holds automatically and KKT are necessary and sufficient for global optimality — for (5) and for each agent problem (16), (17), (18), (19), all of which are convex programs over polyhedra. Multipliers exist because the problem is feasible with finite optimal value. By Theorem 6 the ISO's optimality conditions are exactly the union of the agents' optimality conditions and market clearing, so sufficiency in each agent's problem gives global optimality of that agent's dispatch at λ^*, which is Definition 5. The converse follows by reading Theorem 6 in the ($\Leftarrow$) direction and invoking sufficiency for (5). □

Corollary 8 (No profitable deviation for storage)**.** *At λ^* no storage owner can increase its surplus by deviating, even though it submits no in-horizon energy offer: its intertemporal opportunity cost is represented in the clearing by π_e^* through* (12)*, not by a price forecast.*

Corollary 9 (Individual rationality)**.** *The optimal value of* (18) *is non-negative: $(p_e, g_e, s_{e,t}) = (0, 0, \gamma_e^t s_{e,0})$ is feasible and yields $b_e(\gamma_e^T s_{e,0} - \bar{s}_{e,T})$, which is 0 when $\bar{s}_{e,T} = \gamma_e^T s_{e,0}$.*

Corollary 10 (Revenue adequacy)**.** *The merchandising surplus equals congestion rent and is non-negative:*

$$\mathrm{MS} := \Delta T \sum_{n,t} \lambda_{n,t} \Big[\textstyle\sum_{m,j} d_{m,j,t} + D_{n,t} - \sum_{i,k} q_{i,k,t} - \sum_e (g_{e,t} - p_{e,t}) \Big] = \Delta T \textstyle\sum_{\ell,t} \bar{f}_\ell \big(\bar{\mu}_{\ell,t} + \underline{\mu}_{\ell,t} \big) \ \geq 0. \tag{29}$$

Proof. By (D1) the bracket equals $-\sum_\ell A_{n\ell} f_{\ell,t}$, so $\mathrm{MS} = -\Delta T \sum_t f_t^\top A^\top \lambda_t = -\Delta T \sum_t f_t^\top (\kappa_t - \nu_t)$ by (K1c). Now $f_t^\top \kappa_t = \big(\mathrm{diag}(y) A^\top \theta_t\big)^\top \kappa_t = \theta_t^\top A \,\mathrm{diag}(y) \kappa_t = 0$ by (K1d), so $\mathrm{MS} = \Delta T \sum_t f_t^\top \nu_t = \Delta T \sum_{\ell,t} f_{\ell,t} (\bar{\mu}_{\ell,t} - \underline{\mu}_{\ell,t})$, and by (K7) $\bar{\mu}_{\ell,t} > 0$ only when $f_{\ell,t} = \bar{f}_\ell$ and $\underline{\mu}_{\ell,t} > 0$ only when $f_{\ell,t} = -\bar{f}_\ell$, giving (29). Note MS is exactly the optimal value of the network agent's problem (19). □

Remark 11 (What the equilibrium is an equilibrium in)**.** Theorem 7 is an equilibrium in *revealed* parameters $(\alpha_e, \beta_e, \gamma_e, \epsilon_e, \zeta_e, \bar{s}_e, \underline{s}_e, w_e)$. It is an equilibrium in true values only if w_e equals the owner's true marginal value-to-go $\partial V_e / \partial s_{e,T}$. The design therefore relocates the exercise of market power from a price–quantity offer curve, which an ISO can screen against reference levels, to a declared value-to-go, which it generally cannot verify.

8 Numerical verification

8.1 Case A: the single-node base case of [1]

Section 4.7 of [1] is the specialisation $|\mathcal{N}| = 1$, $\mathcal{L} = \emptyset$ (hence $\Psi = 0$ and all $\lambda_{n,t} = \lambda_t$), $|\mathcal{E}| = 1$, no ramp limits, inelastic demand. Data: $T = 6$, $\Delta T = 4$ h, $\alpha = 0.9$, $\beta = \gamma = 1$, $\epsilon = \zeta = 1$ \$/MWh, $\bar{s} = 800$ MWh, $\underline{s} = 0$, $s_0 = 0$, $\bar{s}_T = 0$, $w = 40$ \$/MWh; demand $(100, 200, 300, 300, 200, 100)$ MW; renewable availability $(200, 100, 500, 500, 25, 0)$ MW at \$0; thermal capacity 100 MW at \$50/MWh. Solving (5) as an LP reproduces the published dispatch, with the charging split between intervals 3 and 4 degenerate (both splits give objective \$19,448.89).

Table 1: Case A: optimal duals (\$/MWh) and dispatch.

t	1	2	3	4	5	6
π_t	49.000	1.111	1.111	1.111	49.000	40.000
$\bar{\nu}_t$	0	0	0	0	47.889	0
$\underline{\nu}_t$	0	47.889	0	0	0	9.000
λ_t	**43.100**	50.000	0	0	50.000	**50.000**
p_t (MW)	100	0	22.2	200	0	0
g_t (MW)	0	90	0	0	100	100
s_t (MWh)	360	0	80	800	400	0

Every condition is satisfied: (K5) $\pi_6 = w = 40$; (K4) $\pi_5 = 40 - 0 + 9 = 49$ and $\pi_4 = 49 - 47.889 = 1.111$; (K2′) at $t = 1$ (interior charging) $\lambda_1 + \epsilon = 44.1 = 0.9 \times 49$, at $t = 3, 4$ $\lambda_t + \epsilon = 1 = 0.9 \times 1.111$; (K3′) at $t = 2$ (interior discharging) $\lambda_2 = 1 + 1.111 + 47.889 = 50$, at $t = 5$ $1 + 49 + 0 = 50$, at $t = 6$ $1 + 40 + 9 = 50$; the idle-direction inequalities hold strictly ($F_t^p = 50, 50, 15 > 0$ at $t = 2, 5, 6$ and $F_t^g = 6.9, 2.111, 2.111 > 0$ at $t = 1, 3, 4$); and (13) for $t = 4 \to t' = 5$ gives $1.111 = 49 - 47.889$, the simplified form (14) not applying because $\bar{\nu}_5 > 0$.

The published price vector is not dual feasible. Reference [1] reports $\lambda = (0, 50, 0, 0, 50, 41)$, arguing that any price in $[0, 43.1]$ clears interval 1 and any price in $[41, 50]$ clears interval 6. That argument tests clearing only against the *offering* resources: renewable at its cap accepts any $\lambda \geq 0$, thermal at zero output accepts any $\lambda \leq 50$. Because the storage submits no in-horizon bid it drops out of that test — but (K2′) applies to it all the same, and $p_1 = 100$ MW is strictly interior, so $\lambda_1 = \alpha\pi_1 - \epsilon = 0.9 \times 49 - 1 = 43.1$ exactly; similarly (K3′)–(K4) force $\lambda_5 = \lambda_6 = 50$. There is no clearing range: perturbing each interval's demand by $\pm 10^{-3}$ MW gives identical left and right derivatives of the optimal value in all six intervals $(43.100, 50.000, 0, 0, 50.000, 50.000)$, so the dual is unique.

Table 2: Case A: storage surplus, best response versus ISO dispatch.

price vector λ	best-response surplus	surplus at ISO dispatch
$(0, 50, 0, 0, 50, 41)$ (as published)	\$76,622	\$51,951
$(43.1, 50, 0, 0, 50, 50)$ (dual feasible)	\$38,311	\$38,311

At the published prices the storage strictly prefers to deviate: it charges 222 MW in interval 1 rather than 100 — energy is free at $\lambda_1 = 0$ — and again in interval 3, discharging 200 MW in intervals 2 and 5. The gap of \$24,671 is a genuine equilibrium failure. At the dual-feasible prices the ISO dispatch attains the maximum exactly, as Theorem 7 requires.

8.2 Case B: three buses, four intervals, with congestion and ramping

To exercise (K1a)–(K1d) and Proposition 3 we solve a networked case. Buses $\{1, 2, 3\}$, lines $\{(1, 2), (2, 3), (1, 3)\}$ with $y_\ell = 1$ and $\bar{f}_\ell = 100$ MW, reference $r = 1$; $T = 4$, $\Delta T = 1$ h. Unit 1 at bus 1 offers two blocks, 100 MW at \$20 and 100 MW at \$40, ramp ± 40 MW; unit 2 at bus 3 offers 300 MW at \$90 with ramp ± 50 MW. Fixed demand $(20, 25, 30, 20)$ MW at bus 2 and $(80, 120, 215, 90)$ MW at bus 3; one elastic block of 40 MW at \$100 at bus 3. One storage unit at bus 3: $\alpha = 0.9$, $\beta = \gamma = 1$, $\epsilon = \zeta = 1$, $\bar{s} = 20$ MWh, $\underline{s} = 0$, $s_0 = \bar{s}_T = 0$, $\bar{p} = \bar{g} = 60$ MW, $w = 30$ \$/MWh. Optimal value \$13,667.22.

Table 3: Case B: primal solution and multipliers.

t	1	2	3	4
$\lambda_{1,t}$ (\$/MWh)	40	40	40	40
$\lambda_{2,t}$	65	65	90	40
$\lambda_{3,t}$	90	90	**140**	40
$\bar{\mu}_{(1,3),t}$	75	75	150	0
$\kappa_{(1,2),t} = \kappa_{(2,3),t}$	25	25	50	0
$\kappa_{(1,3),t}$	−25	−25	−50	0
$q_{1,1,t}$ / $q_{1,2,t}$ (MW)	100 / 60	100 / 62.5	100 / 65	100 / 40
$q_{2,1,t}$ (MW)	0	24.722	60	10
$d_{1,1,t}$ (MW)	40	40	0	40
$f_{(1,3),t}$ (MW)	100	100	100	86.667
p_t / g_t (MW)	20 / 0	2.222 / 0	0 / 20	0 / 0
s_t (MWh)	18	20	0	0
π_t (\$/MWh)	101.111	101.111	139.000	30.000
$\bar{\nu}_t$ / $\underline{\nu}_t$	0 / 0	0 / 0	37.889 / 0	0 / 109

Checks, interval by interval:

- **(K1a), unit 1.** Block 1 is at its cap in every interval, $\bar{\delta}_{1,1,t} = 20$, and block 2 is interior, so $\lambda_{1,t} = c_{1,2} = \$40$ for all t: the marginal block prices the reference bus and no ramp binds ($\varrho_{1,t} = 0$).
- **(K1a), unit 2, and the ramp wedge.** At $t = 3$ unit 2 is interior ($q = 60$, cap 300, ramp slack), so $\lambda_{3,3} = c_{2,1} + \varrho_{2,3}$. The down-ramp at $t = 4$ binds, $\underline{\tau}_{2,4} = 50$, giving $\varrho_{2,3} = -(\bar{\tau}_{2,4} - \underline{\tau}_{2,4}) = +50$ and $\lambda_{3,3} = 90 + 50 = \$140$. A binding *future* down-ramp raises the *present* price by exactly the multiplier. At $t = 4$, $q = 10$ with $\varrho_{2,4} = -50$ and $\underline{\delta} = 0$: $\lambda_{3,4} = 90 - 50 = \$40$.
- **(K1b).** The elastic block is served in full at $t = 1, 2, 4$ with $\bar{\psi} = 10, 10, 60$, giving $\lambda_{3,t} = 100 - \bar{\psi}$; at $t = 3$ it is curtailed to zero with $\underline{\psi} = 40$, giving $\lambda_{3,3} = 100 + 40 = \$140$.
- **(K1c)–(K1d).** $A\,\mathrm{diag}(y)\kappa_t = 0$ holds to machine precision in all four intervals, and $\xi_t = 0$ as predicted.
- **Proposition 3.** With $\Psi_{(1,3),2} = -1/3$ and $\Psi_{(1,3),3} = -2/3$, (10) gives at $t = 3$ $\lambda_2 = 40 + \frac{1}{3}(150) = \$90$ and $\lambda_3 = 40 + \frac{2}{3}(150) = \$140$; likewise $(65, 90)$ at $t = 1, 2$. All six congested-interval prices reproduce exactly.
- **(K2)–(K5).** $\pi_4 = w = 30$; $\pi_3 = 30 - 0 + 109 = 139$; $\pi_2 = 139 - 37.889 + 0 = 101.111 = \pi_1$. Charging is interior at $t = 1, 2$: $\lambda_{3,t} + \epsilon = 91 = 0.9 \times 101.111$. Discharging is interior at $t = 3$: $\lambda_{3,3} = 140 = 1 + 139 + 0$. Idle directions hold strictly.
- **Corollary 10.** MS = \$30,000, and $\sum_{\ell,t} \bar{f}_\ell(\bar{\mu} + \underline{\mu}) = 100 \times (75 + 75 + 150) = \$30{,}000$: identical.

Finally, the equilibrium property itself. At $\lambda_{3,\cdot} = (90, 90, 140, 40)$ the storage's best response to (18) yields surplus \$757.78, exactly its surplus at the ISO dispatch; unit 2's best response to (16) yields \$2,500, again exactly its profit at the ISO dispatch. Both agents have multiple optima — the ISO dispatch is one of them — which is precisely what Theorem 7 asserts and is the generic situation when a resource is marginal.

9 Where the equilibrium breaks

feature	breaks A1 (curvature)	breaks A2 (convex set)
w_e increasing (invalid bid)	**yes**	no
mutual exclusivity via binaries (A.2 of [1])	no	**yes**
minimum charge/discharge levels (A.3 of [1])	no	**yes**
unit commitment, minimum stable levels	no	**yes**
AC network / non-convex losses	no	**yes**
power foldback (A.1.1 of [1])	no	no
SOC-dependent degradation epigraph (A.1.2 of [1])	no	no

- **Non-concave bid.** If w_e increases anywhere, $-b_e$ is non-convex, (K5) is no longer sufficient, and no supporting price need exist. Monotonicity of w_e should be enforced as a bid-validity rule, exactly as monotone offer stacks are for thermal units.
- **Binaries.** With A.2/A.3, or with unit commitment on the thermal side, the feasible set is non-convex, KKT are only locally necessary, a duality gap opens and uplift is required. This is the same pathology as three-part offers, not a storage-specific one.
- **Network non-convexity.** (D2) is the DC linearisation. With a full AC model or non-convex loss representation, Assumption 2 fails, Proposition 3 loses its exactness and the network block of the MCP no longer matches the ISO's.
- **Convexity-preserving extensions.** Foldback is linear; the degradation epigraph $D_t \geq \max\{\cdot,\cdot\}$ is convex; piecewise-linear convex losses are fine. All preserve Theorems 6–7 unchanged.
- **Real time.** Theorem 7 is stated for a fully financially binding horizon. If only $t = 1$ binds and the horizon rolls, the settlement participants face is not the one the KKT system prices, and the multi-interval pricing problem reappears; the day-ahead position then enters the real-time problem as the reference $\bar{s}_{e,T}$.
- **Price selection.** Case A adds a fourth mode, distinct from the others: the model is convex and an equilibrium exists, but the prices used are not the duals of (5). Equilibrium then fails for a purely computational reason — and this is the easiest of the four to repair.

10 Conclusion

Written out in full, the ISO problem prices five distinct scarcities: fuel and opportunity cost through (K1a), willingness to pay through (K1b), transmission through (K1c)–(K1d) and Proposition 3, ramping through the wedge $\varrho_{i,t}$, and stored energy through the costate recursion (11). Its KKT system decomposes exactly into the complementarity conditions of the individual agents plus market clearing (Theorem 6); under a concave end-of-horizon bid and a convex feasible set those conditions are necessary and sufficient, so the ISO dispatch with its duals is a competitive equilibrium supporting no profitable deviation (Theorem 7), storage surplus is non-negative, and merchandising surplus equals congestion rent (Corollary 10). The storage owner never needs a price forecast — but the ISO must use the duals of its own problem as prices, and Case A shows that this is not automatic: on the published base case the reported prices are not dual feasible, the dual is unique, and the resulting misstatement of storage surplus is 64%.

References

[1] R. Baldick. State-of-charge based participation model for energy storage in ISO markets. Source manuscript; equation numbers (1)–(9), Tables 2–4, Sections 4.5–4.7 and Appendices A.1–A.4 cited in this note refer to it.